Ergativity – properly aligned

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Abstract

This is the condensed version of "Ergativity demystified", but supplemented to include causative and applicative constructions.

Ergative languages are described as languages with unusual grammatical features: no ergative Svo languages; optional transitive subjects but obligatory objects; objects agree, even if subjects don't; grammatical voice eliminates objects and thereby makes subjects change into the object's case; subjects are syntactically less accessible than objects, and so on. However, this is a distorted perception. In reality, ergative languages and nom-acc languages follow the same grammatical principles, modulo the reversed alignment of their subjects. The perceptive distortion has its origin in precisely one circumstance, namely the misidentification of the grammatical subject function of ergative languages. Once this is set correctly, the anomalies disappear. What distinguishes the two alignment modes reduces to a single factor, namely the selection criterion for the core argument for the grammatical subject function. The subject of an ergatively aligned sentence is *not* the ergative. The grammatical subject is the core argument that a grammar defines as subject based on grammatical factors.

1. Background

"*Even such commonplace categories as subject and verb are theoretical constructs",* as Gil (2001: 126) notes. "Subject" is a grammatical function based on systematic differences in the grammatical relations and forms of a core argument, but not exclusively on the basis of lexical-semantic content. If these two dimensions – form vs. content – are not kept separate, confusion is unavoidable. This can be clearly seen in grammatical analyses of ergative languages.

The school system has familiarised us with the term "subject", and even AI systems define it in the same way as in school: "*To find the subject, first identify the main verb of the sentence. Then, ask "Who?" or "What?" is performing that action or being described.*" This is also how the grammatical subject of ergative languages is characterised in "Basic Linguistic Theory": "*Subject is simply the association of S, the only core argument of an intransitive clause, and A, that core argument in a transitive clause which could initiate or control the activity.*" Dixon (2010, vol. 1: 229). Remarkably, this is also what Generative Grammar postulates, in different terms of course: Every sentence has a subject in a functional specifier position where it is endowed with its thematic quality as an *agent* (by a hidden something called "little v"). Despite their otherwise so contrasting theoretical viewpoints, both schools insist that ergative is the case of the grammatical subject of a transitive verb, and both schools are united in their error.

Even if this is the majority opinion, it conflicts with too many relevant grammatical facts, as will be shown. What the ergative case marks is not the subject but a 'would-be-subject'. It is the case of a noun phrase that *would be* the subject in a *nom-acc* language. In grammatical terms, the ergative does not qualify as the case of the grammatical subject. The fact that this can be steadfastly ignored is remarkable in itself.

2. Grammatical subject

In linguistics, "subject" is a term used every day. This makes it all the more notable that there is no universally accepted definition. The fundamental difficulty lies in the fact that there are no uniform characteristics across languages that apply equally to every grammatical subject. So, no cross-linguistically valid *absolute* definition of 'grammatical subject' could be achieved. However, a *conditional* definition is sufficient and feasible: The subject is the *grammatically privileged core argument* (see Section 3), as proposed indirectly by Keenan (1976), and directly by Foley & Van Valin (1984), by Mel'čuk (1988:163) and (2014:179), and by Van Valin (1992). The specific way in which the respective privileges manifest themselves depends on the (morpho-)syntactic and structural features of the respective languages.

Furthermore, it makes no sense to refer to a "subject" if in a language there is no core argument of a verb that has mandatory grammatical privileges over the others, see Mithun (2005, 2008) for data and details. In such a language, the notion of grammatical subject is undefined and cannot be meaningfully applied. On the other side of the grammatical spectrum there are languages in which, depending on the grammatical context, a different core argument qualifies as the grammatical subject. Such languages are known as languages with split alignment.

In the majority of languages with morphologically marked grammatical functions of core arguments, a subject is sufficiently well defined by its grammatical status. Note that it is even possible that the respective sets of defining properties for a grammatical subject are disjoint for some languages. In general, however, the characteristics of grammatical subjects overlap considerably across different languages, as the following table shows.

Table 1: Ten grammatical privileges of a privileged core argument

	Table 1: Ten distinguishing grammatical features	Nom-Acc	Abs-Erg
i.	unmarked structural case in case languages	Nom	Abs
ii.	agreement with the finite verb in languages with agreement	Nom	Abs
iii.	indispensability	Nom	Abs
iv.	omission grammatically signalled	Nom	Abs
v.	target of a 'promoted' NP with dependent case	Nom	Abs
vi.	affected by valency increasing (causative, applicative)	Nom	Abs
vii.	top accessibility in languages with accessibility restrictions	Nom	Abs
viii.	target of accessibility advancements	Nom	Abs
ix.	omitted if pronominal in null-subject languages	Nom	Abs
x.	[S[VO]][1]: preverbal argument is the privileged argument	[Nom [VX]]	[Abs [VX]]

2.1 Unmarked case

Greenberg's (1963) Universal 38 states "*where there is a case system, the only case which ever has only zero allomorphs is the one which includes among its meanings that of the subject of*

[1] In [S[VO]] languages, the subject *obligatorily* precedes the VP, which contains the phrase-initial verb and all other arguments. These arguments are postverbal. In SOV and VSO languages, every argument either precedes or follows the verb, respectively. An immediate consequence is the *absence* of the [S[VO]]-typical structural *subject-object asymmetries* in SOV and VSO (Haider 2013: 10-11).

the intransitive verb." Since then, only a small number of languages with marked nominatives – see König (2009, 2023) – or marked absolutives – see Brown (2005) – have been documented. This does not alter the fact that the typical situation is that the unmarked case is nominative or absolutive, whence its name. The term nominative originates from Latin cāsus nōminātīvus, meaning "case of naming". The case for naming in ergative languages is the absolutive. So, let us note that the ergative does not qualify as the unmarked case.

2.2 Subject-verb agreement

Agreement between subject and finite verb is a cross-linguistically widespread morphosyntactic phenomenon. If the verb agrees, it agrees with the subject (e.g. in person and number). If a grammar provides object-agreement for the finite verb it also requires subject-agreement. In nom-acc languages, the finite verb agrees with nominative, in ergative languages it agrees with the absolutive. In his comprehensive study based on twenty Daghestanian languages, for example, Kibrik (1987) emphasizes that the ergative remains invisible for agreement and that the absolutive is the grammatical centre.

In general, if a finite verb agrees with more than one core argument, it always agrees with the nominative or the absolutive, respectively. So, let us note that the ergative does not qualify as the case of the privileged argument.

2.3 Indispensability

As known at least since Keenan (1976), nominative is mandatory while accusative is optional for many verbs, as in "unspecified-object" [2] constructions. The same is true for absolutive and ergative arguments, respectively. Absolutive is mandatory but for many verbs, the ergative core argument may be instantiated optionally, very much like an accusative object. Ergative "subjects" are "dispensable", in Keenan's sense,[3] as described, for example, by Churchward (1953: 68) for Tongan (1a), by Keenan (1976: 313) for Tongan and Tibetan, by Fernández & Berro (2022) for Basque (1b), by Haig (2008, 2018) for Kurmanjî Kurdish ("agentless perfective"), by Haspelmath (1991: 8) for Lezgian, or by Dixon (2010:168) for Dyirbal. He is a rare testimony since he also pays attention to the absolutive: "*In an ergative language like Dyirbal, S and O arguments are obligatory and A optional*." This means, the ergative is optional, the absolutive mandatory and this is a well-known subject-object asymmetry, with absolutive and nominative being the cases of subjects, respectively. In sum, in ergative languages, the absolutive cannot be omitted in the same way as the ergative can be.

(1) a. na'e tamate'i ('e 'tevita) 'a koliate. Tongan (Churchward 1953: 68).
has killed (ERG-David) ABS-Goliath
'David killed Goliath' [without Erg: 'Somebody killed Goliath]
b. (Unai-k) pakete-a-k bidal-i ditu. Basque (Fernández & Berro 2022: 1050)
(Unai- ERG) package-det-pl$_{[abs]}$ send-$_{pfv}$ have.$_{3plAbs[3erg]}$
'Unai has sent the packages.' [without Erg: 'Sb. has sent the packages']

The grammatical source of this property is evident. The core argument with the dependent case (see Section 3) can be omitted. This is the ergative in abs-erg, and the accusative in nom-acc.

[2] He *ate* (*something*). He *left* (some place). He *lost/won* (something). He *married* (someone). He *asks* (something).
[3] "Dispensable" means that a core argument may be omitted. Semantically, the argument is then interpreted by existential closure, that is, as indefinite and unspecific. "*He won*" means that there is *something* that he won.

The privileged arguments, that is, nominative or absolutive, cannot be omitted freely (see the following subsection). So, let us note that the ergative does not qualify as the case of the privileged argument.

2.4 Grammatically signalled omission

In Nom-Acc languages, the *grammatically* permitted omission of the otherwise mandatory, privileged core argument is known as passive. It signals the omission of the core argument that is the candidate for *nominative*. The counterpart in ergative languages is called anti-passive.[4] It signals the omission of the privileged core argument, that is, the candidate for *absolutive* case.

Passive and antipassive are grammatical voices. Grammatical voice is understood here as a general grammatical category of a given language, formally explicitly marked, whose particular value corresponds to operations on the syntactical realisation of argument structures. Antipassive, as characterised by Dixon (1994:146), Dixon (2010, vol.1:166) and Dixon & Aikhenvald (2000:9), is an exemplary case of voice or diathesis in ergative languages:

(2)a. The *antipassive* construction is *formally* explicitly marked.
 b. Antipassive forms a derived *intransitive*[5] from a transitive verb.
 c. The otherwise *ergative*-marked NP becomes S [viz. subject]$_{\mathrm{HH}}$.
 d. The otherwise *absolutive*-marked NP goes into a peripheral function and can be omitted.

Please note that modulo alignment, (2) translates *one-by-one* into nom-acc passive, see (3). In an abs-erg system, the so-called antipassive is grammatically what passive is in a nom-acc system. In fact, antipassive and passive do not deserve to be terminologically separated. They are instantiations of the very same thing, namely the signal of the omission of the argument with default case, but in different alignment systems. “Antipassive” is the passive of ergative languages.

(3)a. The passive construction is *formally* explicitly marked.
 b. *Passive* forms a derived *unaccusative* from a transitive verb.
 c. The otherwise *accusative*-marked NP becomes S (viz. subject).
 d. The otherwise *nominative*-marked NP goes into a peripheral function and can be omitted.

An immediate consequence is the familiar case switch of the argument with dependent case, namely ergative-to-absolutive and accusative-to-nominative, respectively, in the antipassive and the passive of languages with a structural case system. This is the situation in grammars in which a subject is grammatically defined. It crucially does not hold in grammars that do not define a grammatical subject.

The current lack of appreciation for the *immediate* correspondence between passive and antipassive is largely due to a biased perception, namely the confusion of cause and effect. The grammatical cause is the obligatory signalling of the omission of the primary candidate for the default case (which varies with the alignment mode). The effect in nom-acc systems is the elimination of the nominative candidate, linked to the agent of agentive verbs, while in abs-erg

4 *“Ergative systems have an analogous construction, here termed as antipassive, which has all the properties of the passive, as Kuryłowicz, again saw.*” Silverstein (1976: 140), the namesake of ‘antipassive’.

5 The appropriate term for the derived antipassive form of the verb is *unergative* rather than 'intransitive' since the ergative-marked argument of the active construction switches case and surfaces as absolutive.

systems, the eliminated candidate is the absolutive, linked to the 'undergoer' argument.[6] The fact that such effects are used for pragmatic purposes does not make them grammatically causal.

A particular reference to the thematic roles involved is grammatically redundant since it follows from the alignment mode; see Section 3. The cross-linguistically valid generalisation is this. Ergative and accusative are dependent cases and behave alike: When the candidate for the superordinate case – nominative or absolutive – is eliminated, the noun (phrase) with dependent structural case switches into the superordinate case. It is 'promoted'. The promotees are accusative and ergative, respectively, but crucially not the absolutive.

Relation changing is asymmetric. There is no language that 'demotes' a subject to the object function whenever the object is eliminated. In other words, no language turns a nominative into an accusative when the candidate for accusative is absent, and by the same token, no abs-erg language would replace an absolutive by an ergative case when the candidate for the ergative case is absent. The dependency works in the opposite direction, and it is perfectly captured once absolutive is not mistaken for the case of a grammatical object. If it were the case of the object it would be expected to change into the case of the subject, which it does not. So, let us note that the ergative does not qualify as the case of the privileged argument.

2.5. Target of an NP with dependent case that is promoted

The target of 'promotion' is the case of the privileged argument. The licensing/assignment of default structural case has grammatical priority over the assignment of dependent structural case [see Haider (1985a:13, 30); (1986:121-123); (2000: 31, 48); Yip et al. (1987), Marantz (1991), Baker (2015)]. The *dependent* structural case is assigned only in combination with the *default* case. In nom-acc languages, accusative is dependent on nominative; in ergative systems, ergative is dependent on absolutive. If, for instance, the primary candidate for default case is grammatically eliminated, as in the passive/anti-passive constructions, there is a switch from accusative to nominative and from ergative to absolutive, respectively. An inverse switch, that is, from subject case to object case caused by the absence of the object, does not exist, neither in theory nor in reality. In ergative languages, the absolutive is not "promoted" (recte: downgraded) to an ergative when the candidate for the ergative case is omitted, and the same applies to the nominative. The promoted argument is the ergative.

The dependence relation is a push-down-stack relation. The highest in the stack is served first. An immediate consequence is the switch from dependent case to subject case (= default) whenever the default candidate for subject case is absent. In this instance, an argument with dependent structural case becomes the highest argument in the stack and appears as if 'promoted' to the case of the subject. This corresponds to an acc-to-nom shift and erg-to-abs shift, respectively. So, let us note that the ergative does not qualify as the case of the privileged argument.

2.6 Affected by valency increasing operations (causative, applicative)

Causative and applicative constructions are typical examples of how the *same* case-dependence relations lead to *different* surface patterns due to the different alignment mode. In both

[6] The functionalist perspective, cf. Cooreman (1994), focuses on the effects, that is, the contexts of use, such as patients as unknown, unimportant, or less important than the agent, and disregards the primary grammatical question: Why not simply omit the patient argument, on a par with the ergative, and on a par with nom-acc languages?

constructions, an argument is added to the core arguments of the base verb. In a causative construction, the added argument denotes the causer. This qualifies it as the top argument in the ranking of arguments in the argument structure and therefore, it is assigned the case the top argument of the base verb would receive. In applicative constructions, a core argument is added that qualifies as a candidate for the bottom position of the argument structure and therefore as a candidate for the case that the lower, structurally cased argument of the base verb would receive. Table 2 and (4) illustrate the respective effects. The subscripts C and A stand for the added core argument in the *causative* (C) construction and in the *applicative* (A) construction. In Nom-Acc *causatives,* the original candidate for nominative becomes accusative (Table 2, line a). In Abs-Erg, it is the *applicative* construction that triggers the switch (Table 2, line b), for the same reason. Under each alignment mode, a case switch is triggered *only* when the case that the added argument gets assigned is the default case, that is, nominative or absolutive.

Table 2	Nom-Acc			Abs-Erg	
a. Causative	Nom_C + Nom	⇒ Nom_C – Acc		Erg_C + Abs	⇒ Erg_C – Abs
b. Applicative	Nom + Acc_A	⇒ Nom – Acc_A		Abs_A + Abs	⇒ Abs_A – Erg

Causative and applicative shift the intransitive format of the base verb to the canonical transitive format. When the base verb is transitive already, things get complicated because there are several options. Let us go over relevant details.

Causative is defined as follows by Dixon (2010, 3: 467): "*Causative: valency-increasing derivation which prototypically operates on an intransitive clause, putting underlying S argument into O function and introducing a 'causer' as A argument.*"

This characterisation implies a consequence with different effects, depending on the alignment mode (see Table 2). In Nom-Acc languages, the causer-argument is a candidate for nominative, that is, the default case. So, the nominative of the base verb is fitted into the format of a transitive verb. It is 'demoted' to the dependent case, viz. Accusative, as in German (4b). If the base verb is transitive, the result is a construction with double accusative (4c).

(4) German
   a. Der_{Nom} Druck sinkt.
      the pressure lowers
   b. Die Maßnahme senkt den_{Acc} Druck.
      the measure lowers the pressure
   c. Er lässt den_{Acc} Lehrling den_{Acc} Druck senken.
      he has/makes the apprentice lower the pressure.

In an ergative language, just like in nom-acc languages, the causer-argument corresponds to the agentive core argument of a transitive verb, but it is the candidate for dependent case, that is, ergative. So, the absolutive case of the other argument of the base verb is not affected by the transition to a transitive format in the causative version. A telling difference shows only when the base verb is a transitive verb. In ergative languages, the result can be a verb with *two* candidates for the structurally dependent case (5). This is the exact parallel to (4c) in nom-acc, with two accusatives.

According to Dixon (2010, vol. 3: 256), we may see several outcomes across languages, but he does not differentiate between the alignment modes: "*Original A retains A-marking*" or

"*Original A has O-marking.*" For a transitive verb, the original A is ergative under ergative alignment. If it retains this case, the result is a double-ergative construction, as in Trumai (5), from Dixon (2010, vol. 3: 257).

(5) Trumai

Alaweru-k	hai-ts	axos	disi	ka
Alaweru-ERGATIVE	1sg-ERGATIVE	child:ABSOLUTIVE	beat	CAUS

Alaweru made me beat the child

Since the ergative is the dependent case, the doubling corresponds to the doubling of accusative in a causative construction in nom-acc languages as for instance in German (4c). In languages that do not tolerate doubling, the case of the nominative candidate of the base verb shifts to the case of an indirect object, as for instance to dative. (6) illustrates this with data from French (6a,b), for nom-acc, and Svan (6c,d) for abs-erg.

(6) French

a. Jean chante la Marseillaise.
   Jean sings the M.

b. Elle fait chanter la Marseillaise à Jean.
   she makes sing the M. $\text{to}_{\text{Dat.}}$ Jean – 'She makes Jean sing the M.'

Svan [Dixon (2010, 3:264)]

c.

dena-d_A	kalaxwem	mare-s	diar_O
girl-ERGATIVE	give:AORIST	man-DATIVE	bread:NOMINATIVE

The girl gave bread to the man

d.

eǯe-m_A	kalaxawodnune	dena-s
he-ERGATIVE	give:CAUSATIVE:AORIST	girl-DATIVE
diar_O	mare-š-t'	
bread:NOMINATIVE	man-GENITIVE-FOR	

He made the girl give bread to the man

What is grammatically barred under both alignment modes, however, is a doubling of the default case, that is, two core arguments with nominative or absolutive, respectively.

Let us turn now to ***applicative***. According to Dixon (2010, 2: 423), applicative is a "*valency-increasing derivation which prototypically operates on an intransitive clause, putting underlying S argument into A function and introducing a new O argument (which may have been in peripheral function in the underlying clause).*"

Structurally speaking, not in terms of content, the applicative construction is the flip-flopped version of the causative constructions. The core argument introduced and added to the base verb is an argument dedicated for the case of the *lower* ranked structurally case-marked argument in the lexical argument structure of a verb. This is an argument with accusative (= *dependent* case) in the nom-acc mode but with absolutive (= *default* case) in the abs-erg mode. This makes the essential grammatical difference.

The applicative construction, as well as the causative construction, introduces a new core argument which interferes with the case assignment conditions of the base verb *whenever* the new argument is assigned the case of the privileged argument. The case dynamics are the same, only the contexts in which they take effect are different. In the nom-acc mode, it concerns the nominative candidate of the base verb (see causative) while in the abs-erg mode it concerns the absolutive candidate (see applicative).

In the following example from Dyirbal, the absolutive candidate of the base verb (7a) is assigned dative in (7b), which mirrors nominative-to-dative in French (6b). In (7d), the absolutive added to (7c) makes the absolutive argument of the base verb shift into the dependent case, i.e. ergative. This is the parallel to the nom-acc shift in the nom-acc causative construction. (7f) illustrates that in a nom-acc language, an applicative construction does not affect the nominative since it introduces an argument for dependent case.

Dyirbal

(7) a. baŋgul$_{A}$ gaban$_{O}$ gunda-n jawun-da
HE:ERG grub:ABS put.in-PAST dilly.bag-LOCATIVE
He put the grubs into the dilly-bag — Dixon (2010, VOL.3: 297)

b. baŋgul$_{A}$ jawun$_{O}$ gundal-ma-n gaban-gu
HE:ERG dilly.bag:ABS put.in-APP-PAST grub-DATIVE
He put the grubs into the dilly-bag (lit. He put-in grubs to the dilly-bag')

c. bayi$_{S}$ Nurba-ñu <u>barrmba</u>-ba$_{S}$ — Dixon (2010, vol. 3: 297)
he$_{Abs}$ return-*past* quartz-with
'He returned with the quartz'

d. baNgul$_{A}$ barrmba$_{O}$ Nurbay-ma-n
he$_{Ergative}$ quartz$_{Abs}$ return-*applicative-past*
'He returned-with the quartz'

German

e. Er$_{Nom}$ arbeitet (an einer Anfrage).
he works (on a request)

f. Er$_{Nom}$ *be*arbeitet eine Anfrage$_{Acc}$.
he *be*$_{Appl}$-works (=handles) a request

The outcome of this comparison of causative and applicative modulo alignment is what is summarised in Table 2: Whenever the respective privileged arguments – nominative or absolutive – are directly involved, the very same grammatical case-shift mechanisms apply. Nominative and absolutive behave grammatically in parallel in their respective syntactic contexts. So, let us note that the ergative does not qualify as the case of the privileged argument.

2.7 Top accessibility in languages with accessibility restrictions

A reliable indicator of a syntactically privileged argument function is Keenan & Comrie's (1977, 1979) accessibility hierarchy (AH). No languages are known in which an undisputed grammatical subject would be inaccessible for relative clause formation, but there are languages in which *only* the subject is accessible. Dixon (2010, 2:220) notes that "*a fair number of languages with an ergative orientation allow the CA to be in just S or O (not A) function within the RC*." Polinsky et al. (2012: 69) acknowledge that "*ergative languages have posed challenges to the AH in that many of them exhibit syntactic ergativity: In many of them, the absolutive arguments (intransitive subject and transitive object) relativize with a gap, but the ergative*

DP does not." Evidently, the absolutive behaves like the nominative and unlike an object. So, let us note that the ergative does not qualify as the case of the privileged argument.

2.8 Accessibility advancements

The accessibility restrictions in terms of the accessibility hierarchy are at the same time a test criterion for grammatical privileged arguments of a given languages. "*Languages which allow a limited set of functions for a common antecedent in relative clauses or in main clauses (or in both) are likely to employ syntactic derivations which will put an argument originally in a non-CA-allowed function into one that is permitted*." (Dixon 2010, vol. 2: 324). For ergative languages, this means the following: "*a major function of antipassive is to place the underlying A argument into derived S function in order to satisfy some S/O condition within the grammar*." Dixon (2010, vol. 3: 217). Haude (2024: 8) refers to the same function of antipassive in Movima: "*It has an exclusively syntactic function: It is only used for extraction (i.e. relativization, referentialization, topicalization*."

These are perfect descriptions of the *effect* and the *direction* of advancement to the case of a grammatically privileged core argument. The target of advancement is the absolutive case (or its structural equivalent). There is no clear case of a language that would demote a subject to an object in order to meet the demands of accessibility. So, let us note that the ergative does not qualify as the case of the privileged argument.

2.9 Omitted in null-subject languages if pronominal

A null-subject language is a language that allows the privileged argument to be omitted in finite clauses if it would be pronominal. The omitted argument is interpreted like an overt personal pronoun. In nom-acc languages, the omitted pronoun is the candidate for nominative. In a prototypical ergative null-subject language (without split for pronouns), the omitted pronoun corresponds to the argument with absolutive case. It is difficult to obtain clear data for this feature, as ergative language in which the finite verb only agrees with the absolutive and at the same time only the pronouns with which it agrees can be omitted, are rare. Brown (2001: 242) discusses pertinent data from Nias. However, the reverse conclusion is revealing. Unlike nom-acc languages, there is no known ergative language in which only an ergative pronoun is omitted.

The null-subject property of pronouns must not be confused with either topic-drop or optional omission of an argument. In a topic-drop language, not only subjects but also objects can be omitted if they are topics. A null-subject language is a language in which *only* pronominal *arguments* are omitted if they agree. German, for instance, is not a null-subject language (8a), but drops pronouns with structural case (nom, acc) in the clause-initial position of main clauses (8b,c), that is, subjects or direct objects.

(8) German
  a. Gerade hat *(er) angerufen und gesagt, dass *(er) nicht teilnehmen kann.
     just has (he) called and said, that (he) not participate can
  b. [Wo ist der Hund?] (Der_{Nom}) sitzt schon auf dem Sofa.
     Where is the dog? (It) sits already on the sofa.
  c. [Wo ist der Hund?] (Den_{Acc}) wirst du gleich hören.
     Where is the dog? (It) will you in-a-moment hear

Finally, the optional omission of an ergative (see Section 2.3), does not qualify as a null-subject property. It is an instance of omitting an argument with dependent case. In this case, the missing argument is interpreted as an implicit argument, that is, as indefinite and unspecific. The null subject of a null-subject language is interpreted as a personal pronoun, that is, definite and specific and as anaphoric in suitable contexts.

2.10 [S[VO]]: preverbal argument as the grammatically privileged argument

The bracketing in the section title indicates the essential property. It signifies that in such languages, there is a unique position for the subject, which precedes a constituent with the verb at the beginning, followed by the remaining arguments. In an [S[VO]] language, this clause-initial position is obligatory for subjects. A typical sample of such languages contains English, North-Germanic, and Romance languages. In (Greenbergian) word order typologies, "SVO" is used differently. It is a conflation of genuine [S[VO]] languages and free word order languages in which S-V-O is a frequently used pattern. An example of the latter type are Slavic languages: "*In each of the Slavic languages all twenty-four possible combinations of a subject, direct object, indirect object and verb occur as grammatical declarative orders.*" Siewierska & Uhlířová (2010:109). Nevertheless, the S-V-O order is frequently encountered for pragmatic reasons, which is why Slavic languages are classified as SVO; see Haider & Szucsich (2022).

[S[VO]] languages and languages with free word worder with a preference for S-V-O serialisation are different types.[7] The customary use in typologies conflates the two types under "SVO", as can be seen from the following definition of "dominant word order": "*This means that it is either the only order possible or the order that is more frequently used.*" (Dryer 2013). Obviously, this definition joins two grammatically disjoint properties, namely a strict word-order property (viz. "the only order") with a variable-word-order property (viz. "more frequently used"). In the strict-order type, order variation is ungrammatical, for grammatical reasons. Two disjoint types must not be contaminated because the result would be inconsistent. As a consequence, the currently used category "SVO" does not make reliable predictions of correlations. It was precisely for this reason, that Hawkins (1983: 114-16) abandoned SVO as a typological indicator. [S[VO]], on the other hand, leads to a multitude of welcome implications, as shown in Haider (2015a) and Haider & Szucsich (2022), but it is more difficult to identify.

With this in mind, let us turn to the central topic of this subsection. Although the "SVO" type is one of the major word order types, there are allegedly no ergative "SVO" languages. (viz. Erg-V-Abs), but numerous languages are attested as ergative "OVS" (that is: Abs-V-Erg), a virtually inexistent word order type under nom-acc alignment. An [Acc [V Nom]] language has never been discovered.

Typologists [Trask (1979), Nichols (1992), Siewierska (1996)] put the supposed gap on the list of properties of ergative languages. Generativists [Mahajan (1997), Lahne (2008), Taraldsen (2017), Roberts (2021)] try to reconcile Generative Grammar with this alleged fact. For these scholars, "*no ergative SVO*" means that there are no languages with an [Erg [V Abs]] clause structure. However, this is *not* the clause structure of an ergative [S[VO]] language. Ergative SVO clauses are [Abs [V Erg]], and they exist, misfiled as OVS (see Sect. 4). On the other hand, it is almost trivially true that [Erg [V Abs]] languages must be inexistent; there are no

[7] *"Languages with highly flexible word order are themselves a linguistic type."* (Dryer 2007:113).

[Acc [V Nom]] languages either. An argument with dependent case in the typical structural subject position of an [S[VO]] clause structure plus the argument with default case in a VP-internal position is no admissible canonical base structure of clauses in any language.

Languages with [Abs [V ERG]] clause structure have been (mis-)classified as OVS languages without any objection. They need to be re-classified as "SVO" languages (see Sect. 4). The preverbal, non-agentive noun phrase in an ergative “OVS” language is not the syntactic object. It is the syntactic subject of an abs-erg-language. This subtle point may be harmless for the linguistic description of an individual language but the subsequent typological interpretation, that is, the step from "agent" or "patient" to "subject" and "object", respectively, is detrimental if “ergative” is equated with “subject case”.

In other words, there is no justification for classifying an *ergative* language as "OVS" whenever its obligatory serialization pattern in simple clauses with non-pronominal noun phrases happens to be [Patient [V Agent]]. This, however, is exactly what happens in typological surveys. For instance, in WALS (Dryer & Haspelmath 2013), Päri is listed as "OVS" (feature 100A) and "ergative" (feature 81A) although in an earlier publication, Dryer (2007: 70) himself has explicitly qualified such a classification as “*somewhat misleading*.”[8] So, let us note again that the ergative does not qualify as the case of the privileged argument.

2.11 Interim result

In the preceding subsections, ten characteristics are presented as evidence for the status of “grammatically privileged argument,” which is nothing more than a paraphrase of “grammatical subject.” If a grammarian is asked whether a core argument with the nominative case is considered a grammatical subject if it fulfils the characteristics listed in Table 1, the answer cannot be “no.” For the very same reason, one cannot deny at the same time that the absolutive is the grammatical subject of an abs-ergative language. Nominative and absolutive behave grammatically parallel in the corresponding grammatical contexts. If the nominative case is the case of the grammatical subject, then the absolutive case is also the case of the subject. The only relevant difference is the mode of alignment.

2.12 Ergative as subject case? – An understandable misunderstanding

The assumption that the ergatively marked noun or noun phrase is the subject is a misinterpretation with a long history – e.g. Fabricius (1801:78),[9] Uslar (1889), Schmidt (1902); see Vollmann (2008: 129) for a historical survey – rooted in a centuries-old grammar tradition. It originated at a time when Latin was the grammatical model for describing other languages. The terminology was descriptive, without explanatory ambitions. Subject is what corresponds to the subject in a Latin clause when translated into another language. There was no differentiated theory of grammar at that time. The second[10] source of the misunderstanding is Eurocentrism; see Gil (2001). Ergative languages appeared to be exotic compared to Indo-European

[8] Dryer (2007: 70): "*Characterizing such languages as OVS is somewhat misleading in that the word order really follows an ergative pattern Abs-V-(Erg).*"

[9] He refers to the ergative as *„nominativus transitivus“* (‘transitive nominative’).

[10] There is a non-linguistic reason, on top of it: “*People tend to follow one another in what they say, do, and think. If there is a habit of doing things in a certain way, then people will do them in this way. So it is with language description.“* Dixon (2010, vol.1: 66).

languages.[11] From this perspective, it was possible to maintain the biased perception that the grammatical subject of a typical transitive verb is universally associated with the role of the agent. The ergative would be the exotic subject case, even if the absolutive had to be downplayed to an exotic object case, or an exotic subject case of intransitive verbs, since it cannot be denied that it is the case of the subject of intransitive finite verbs in ergative languages.

2.13 Subject properties of an ergative argument?

A treatise on the subject qualities of the absolutive is incomplete without addressing the alleged subject characteristics of ergative noun phrases. Anderson (1976) has pointed out, and Dixon (2010, vol. 1: 229)[12] elaborates on it, that the ergative-marked argument can appear in contexts which are the domain of nominative subjects in nom-acc languages, such as imperatives, the antecedents of reflexives, or as controlled subjects in infinitival constructions. That is correct, but not compelling. These functions are not exclusively reserved for grammatical subjects.

Let us start with ***imperatives***. As they are used to issue *direct* commands, requests, or instructions, the addressed participant is the person to whom the agent argument of the verb refers. In nom-acc languages, this coincides with the subject argument. But this is only an accompanying feature. The essential property is the reference to the addressed participant with the agent role of the verb. If the grammar of an ergative language required the imperative to be related to the absolutive, the message would not be conveyed to the intended recipient. Imperatives with the absolutive argument of a transitive verb as referent would be dysfunctional. In nom-acc languages as well as in abs-erg languages, the addressee is the relevant argument role for imperatives, irrespective of the subject property. For functionalists, this ought to be self-evident. Interestingly, their argument here is structural rather than functionalist. We can trust that the evolution of grammars would not have enforced pragmatically dysfunctional imperatives.

As for **reflexives**, they are typically bound by subjects in a nom-acc grammar. However, this is not the only option. Reflexives may be bound also by the argument with dependent case, that is, the accusative, as illustrated by English (9a,b) and German (9c,d).

(9) a. [Circumstances don't make the man.] They only reveal *him* to *himself.*
b. When squaring, you multiply *a number* by *itself.*
c. Die Krankheit hat *sie* mit *sich selbst* konfrontiert.
the illness has her with herself confronted
d. Schwerhörigkeit ließ *ihn sich* aus der Öffentlichkeit zurückziehen.
hearing-loss made *him himself* from the public withdraw

What this examples show is that an argument with a dependent case is in principle a licit antecedent of a reflexive if the antecedent c-commands it. So, an ergative core argument is not excluded as an antecedent of a reflexive. But why would the agent argument be a preferred argument? As in the case of imperatives, cognitive conditions seem to be responsible for this.

[11] "*This opinion was assisted by the fact that most ergative languages then known were spoken by small communities outside the mainstream of European culture (even the Basques were characterised in this way).*" Dixon (1994: 214).
[12] "*Even in ergative languages, S and A share a number of properties – as addressee in imperative constructions, as controller of reflexive, and so on*".

Cognitively, the semantics of (agentive) transitive verbs establishes a causal relation. An agent does something and the one affected by this action is identical with the agent. If in an ergative grammar, the participant that binds the reflexive pronoun is the ergative, then the agent of the situation is directly linked, just as in nom-acc languages. Linking of a reflexive via the absolutive would be a grammatically not enforced cognitive detour in many cases. Here is an example.

(10) a. Donald refutes himself.
b. [REFUTE (x,y)] & x = y (i.e. reflexive): [REFUTE (x,x)]
c. ⟦Donald⟧ = d. ⟦REFUTE (d, d)⟧

Let us assume the absolutive argument were the *only* licit antecedent of a reflexive. Then the ergative noun phrase must satisfy the variable corresponding to the reflexive. In this case, we would be told by an equivalent of (10b) that (10a) tells us that something happened to y = $\text{Donald}_{\text{Abs}}$, namely that someone refuted Donald. Only then would the reflexive indirectly provide information about the agent, namely "himself".[13] The direct way with ergative as the binder seems to be the cognitively straightforward one. If an ergative serves as antecedent of a reflexive, this is not evidence for the subject status of ergative, contrary to what Polinsky et al. (2012: 268) emphasise, but only a parallel to nom-acc in terms of the semantics of the involved arguments. In each case, the grammar exploits a grammatically licit way of linking the reflexive to its antecedent. In sum, neither imperatives nor reflexives provide compelling evidence for the subjecthood of ergatives in ergative languages.

Finally, ***infinitival constructions*** have been adduced as evidence for the subject status of ergative. One typical argumentation goes as follows. If the main verb takes an infinitival complement, "*the two verbs must have the same subject (S or A) irrespective of whether the language is accusative or ergative*". Dixon (1994: 135). Consequently, the ergative must be a subject. The other argumentation focuses on the potential presence of an absolutive argument in infinitival complements. This is taken to be evidence against the subject status of absolutive case, given that the infinitival subject would have to remain silent in an infinitival complement. Neither of these arguments is compelling, however.

Dixon's argument would be compelling only if ergative were the *only* admissible case for the silent subject of an infinitival subject in a finite clause. This is not true, of course, since "S" is absolutive and intransitive infinitival complements are frequent. So, the relevant question is "Absolutive and what else?" Haspelmath (2019:122) points out that in Lezgian, a silent argument of an infinitival complement may represent an *absolutive*, a *dative*, or an *ergative* argument, depending on the infinitival verb. Icelandic is the nom-acc counterpart, as Sigurðsson (2008: 412) illustrates. There are infinitival constructions with a silent nominative, dative or inherent accusative as silent arguments which may even trigger case agreement with a predicative adjective. Ergative is merely a permissible non-subject candidate alongside the absolutive.

An example of the other kind of argumentation is the following. Polinsky (2017: 17) notes that "*many ergative languages, including Inuit, have the absolutive freely available in non-finite clauses*" and that "*the so-called contemporative form of the verb in Inuit/Inuktitut is characterized by agreement with the absolutive but not the ergative*." The second part of the quote is the

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[13] This is grammatically admissible but not preferred. A Google books (20th cent.) search produced 10^6 hits for "*has refuted himself*", but only seven hits for "*has been refuted by himself*" (Feb. 2026).

key to the first, namely agreement. The very same constellation can also be found in nom-acc languages such as Portuguese, with nominative occurring in the so-called inflected-infinitive construction, cf. Madeira & Fiéis (2020). Another example is Hungarian, cf. Tóth (2020). In each case, agreement licenses the assignment of subject case, in finite as well as in infinitival context, since both contexts provide an inflected verb with agreement features. The typical infinitival clause without a lexical subject is a clause without subject-verb agreement. Lack of agreement (in languages with subject-verb agreement) is the factor blocking the case of a syntactic subject. If the 'infinitival' verb can agree, the absolutive is not barred from an infinitival complement.

In summary, it can be stated with certainty that imperatives, reflexivisation, and infinitival complementation are not clear distinguishing features for grammatical subjects. These contexts do not exclude non-subjects, especially those with structural case, viz. dependent case. Even if an ergative may be involved in these constructions in a similar way as a subject is involved otherwise, this is no conclusive evidence for its status as a grammatical subject.

3. Subject, subject selection and alignment

For illustrative purposes, let us assume that the selection of the subject candidate were part of a programming task to develop an efficient and economical system for linking arguments. How should we proceed? We need two *structurally* assignable cases, one of which is the default case. Second, we establish a dependency relation between default and dependent case so that the default case (which feeds agreement) is instantiated also if its primary candidate is omitted. What we end up is the following situation (11):

(11) a. V (A) default case (intransitive; unaccusative/unergative)
b. V (A_1, A_2) A_1 or A_2 with default case?

The argument of an intransitive with structural case (11a) defines the subject (= default) case. In (11b), the prototypical constellation of a transitive verb, there are two candidates with structural case. One has to receive the default case, making the other argument the candidate for the dependent case, viz. accusative and ergative, respectively. But which one? Logically, there are two possibilities for the choice of the candidate, namely A_1 or A_2. Consequently, there are two options for the choice of the candidate for default case. This is the alignment parameter in Figure 1. The resulting *alternative* alignment modes are known as the nominative-accusative mode and the absolutive-ergative mode.

Figure 1: Alignment parameter. See Haider (1985a), (1985b), (1986), (2000).

Lexical argument tier: $[\Theta_{AG} < \Theta_{EXP} < \Theta_{TH}]$ (agentive di-transitive verb)

Syntactical argument tier: A_1 A_{OBL} A_2 (OBL = oblique = inherent case)

a) A_1 with default case = Nom-Acc alignment ⇒ Subject: A_1
b) A_2 with default case = Abs-Erg alignment ⇒ Subject: A_2

Let us note that the choice of candidate for the subject case (= subject) is structurally based. In the lexical argument structure of verbs, the arguments are ranked. The nom-acc mode defines the highest structurally cased argument A_1 as subject candidate. The abs-ergative mode defines the lower one, i.e. A_2, as subject. The rest follows and works in the same way in both systems.

In both cases, the semantic content of the respective argument is secondary. That in the nom-acc mode, the agent argument of a verb is always selected as subject is merely a side effect of the ranked argument structure (which follows from the semantic decomposition). The agent argument is the highest in the ranking of arguments. On the other hand, the argument slot for the undergoer is the lowest in the ranking of core arguments; see Van Valin & LaPolla 1997: 177). Experiencers are in between.

The *dependent structural case* is the case that is assigned to the remaining argument with structural case. It is dependent on the subject case as it is the option that is only selected if the default option is also implemented. Let us briefly contemplate two implications with empirically testable consequences. An immediate consequence is that two different argument structures of intransitive verbs necessarily end up with the same case configuration, namely (12a) and (12b). In each alignment mode, the structurally-cased argument of (12), that is A_2 in (12a) and A_1 in (12b), will end up in the default case.

(12) a. V [(A_{OBL}), A_2] *unaccusative* verb in a nom-acc language | intransitive in abs-erg
b. V [A_1, (A_{OBL})] *unergative* verb in an abs-erg language | intransitive in nom-acc

In (12a), in a nom-acc language, A_2 corresponds to the dependently marked argument (= accusative) of a transitive verb. In an ergative language, however, it qualifies as an intransitive[14] verb with the canonical candidate for the subject function. Analogously, in (12b), A_1 corresponds to the dependently marked argument of a transitive verb in an abs-erg language. On the other hand, (12b) is the canonical format of an intransitive verb in nom-acc. For nom-acc languages, it is known since Perlmutter (1978) that there are two classes of monovalent verbs, namely intransitive proper (12b) and unaccusative verbs (12a). Their subject is an argument that would receive the dependent case in a transitive context. As a single argument of a verb, it is 'promoted'. Therefore, it shares properties with the passive subject of a transitive verb.

Under these circumstances, we have to expect a corresponding situation in ergative languages. Verbs with the argument structure (12a) are verbs with a canonical candidate for the absolutive subject. However, the absolutive subject of verbs of class (12b) – unergative verbs – is the candidate for the dependent case in a transitive context, namely the ergative case, but as a single argument it is 'promoted', as in an antipassive construction.

Another welcome implication of the alignment parameter is the possibility of a *dual*-alignment system, known as "split" system.[15] Option (a) and option (b) of Figure 1 can coexist in the same grammar, in separate contexts, though. Here is an example: Dixon (1994: 177) concludes his discussion of the grammatical pivot of coordination in Yidin^{y} as follows: "*It would be nice if one could uncover a universal rationale for the occurrence of S/A and S/O pivots in a single language.*" The solution is the alignment parameter.

If a language has a split alignment system with aligning *non-pronominal* core arguments by

[14] Intransitive = The single core argument with structural case is the default candidate for subject. The single argument with structural case of an *unaccusative/unergative* verb is the default argument for dependent case,

[15] "Split" is not an apt metaphor. It suggests that something is broken up. In fact, nothing is "split". There is only a *differentiated* mode of alignment, contrasting with *uniform* alignment in other languages.

abs-erg and *pronominal* ones by nom-acc, this results in two different contexts of qualifying for a grammatical subject. Yidin^y is such a language, see Dixon (1994: 176). First, "*nouns inflect on an ergative pattern and pronouns on an accusative one* (*1st and 2nd person*)". Second, "*Yidin^y has an S/O pivot for all types of relativisations and one kind of coordination, but an S/A pivot for another kind of coordination.*" The "*another kind of coordination*" is one with a pronominal antecedent: "*There is an S/A pivot for joining together two clauses which have a common NP that is pronominal, and an S/O pivot for linking clauses whose common NP is non-pronominal.*" Pronominal means personal pronouns. So, the *grammatical subject* is S (= Nom) and A (= Nom) for pronominals, but S (=Abs) and "O" (= Abs) for non-pronominal arguments, and in each case, the pivot is grammatically identical, namely the *grammatical subject* (with absolutive and nominative as subject cases in alternative contexts). Relativisation works without pivot split because personal pronouns (1st and 2nd person) are no target of relativisation.

This is an on-target showcase of the explanatory power of an empirically adequate concept of "grammatical subject". Dixon's pivot of coordination reduction is the *grammatical subject* in a dual alignment system[16] (viz. abs-erg & nom-acc). Dual systems have different contexts for defining grammatical subjects, namely one in an absolutive-ergative mode and another one in a nominative-accusative mode. Consequently, such a language has more than one type of pivots, either the absolutive in abs-erg contexts or the nominative in pronominal nom-acc contexts, in the very same language.

4. "OVS" = SVO under ergative alignment

This section[17] aims to document that almost all of the (un)disputed candidates for the category "OVS language" are Abs-V-Erg languages, with the so-called undergoer-argument as syntactic subject. Therefore, an ergative "OVS" language, when classified correctly, is an SVO language with abs-erg alignment. The cause of the present misperception is the non-structural characterisation of grammatical functions, namely the equation of a lexico-semantic stereotype, viz. agentivity, with grammatical subject status.

In most typological surveys the word order P-V-A is classified as "OVS", without taking into account the particular alignment system of the given language, which is almost always ergative. Dixon (1994: 50) explicitly notes that for "*languages with syntactic function shown by constituent order*", SV/OVA is a sign of ergativity. S in "SV" and O in "OVA" stand for absolutive.

Greenberg (1963: 76) described OVS as one of the types that "*do not occur at all or, at least are excessively rare*", and this has proven correct, contrary to positions held in the typological literature. Greenberg's (1963) original sample of thirty languages contained only two languages classified as OVS, with VOS as alternative word order,[18] namely Siuslaw and Coos [s. Greenberg's (1963) Appendix II]. Both languages are ergative; see Mithun (2005), Frachtenberg (1913: 128, 154).

Hawkins (2014:189) takes up an observation made by Primus (1999) and emphasises that the majority of object-initial languages are ergative, "*including OSV languages (Dyirbal, Hurrian,*

[16] As typologist have documented, dual alignments can occur in several grammatical dimensions: case-assignment or subject-verb agreement, partitioned by noun (phrase) type, verb type, tense, etc.
[17] This section is a slightly extended version of Section 3 of Haider (2021a).
[18] Interpreted properly, this confirms Greenberg's Universal 6: SVO as an alternative word order of VSO.

Siuslaw, Kabardian, Fasu) and OVS languages (Apalai, Arecuna, Bacairi, Macushi, Hianacoto, Hishkaryana, Panare, Wayana, Asurini, Oiampi, Teribe, Pari, Jur, Luo, Mangaraya)." In a survey, Hammarström (2016: 25) calculated the constituent orders of 5252 languages partitioned into 366 language families. His count results in 40 OVS languages belonging to three languages families.

Dixon (1994: 50-52) itemizes the following ergative languages as instances of SV/OVA, that is, *ergative* "OVS" languages: Kuikúro, Macushi,[19] Maxakalí, Päri, and Nadëb. Further confirmation can be found on Kuikúro in Franchetto (1990, 2010), on Macushi in Abbott (1991) and Carson (1982), on Maxakalí in Popovich (1986), on Päri in Andersen (1988), and on Nadëb in Martins & Martins (1999). Dixon also refers to a second pattern, namely VS/AVO, and illustrates it with Huastec and Paumarí. This deserves a comment, since from the perspective favoured in this article, "AVO" would mean ergative-V-absolutive, and therefore, O-V-S in an ergative setting.

Huastec is described as an SVO language by Edmonson (1988), but Kondić (2012: 283) illustrates the flexible word order of South Eastern Huastec – SVO, OVS, VSO – with the order variants of a declarative clause and notes that "*the free placement of the subject and object in (50) is possible partly because of the 'natural state of the things.'*" Since Mayan languages are predominantly V-initial (England 1991), the Huastec data do not provide compelling evidence for a *basic* [O[VS]] structure. XVS is the result of fronting X in a basic VSO clause.

Chapman & Derbyshire (1991:164) classify Paumarí as ergative and "SVO" but this is only half of the relevant information, since it employs the nom-acc mode at the same time. According to Derbyshire (1983: 24) "*the ergative and accusative constructions can be used interchangeably, and the choice of one over the other is arbitrary, or just a matter of style*." Dixon (1999: 305) notes that the marked noun phrase – 'ergative' or 'accusative – is typically clause initial. Zwart & Lindenbergh (2021: 30) argue that when the respective clitics are used as 'case' markers, viz. only in the preverbal position, then S, A, and O, respectively, are coded differently, which is a tripartite system. So, Paumarí does not qualify as an ergative language and consequently it is syntactically not an [O[VS]] language.

In a study on word-order type and alignment, Siewierska (1996) lists four languages as "OVS" out of a sample of 237 languages, namely Macushi and Päri, as in Dixon's sample, plus Hixkaryána, and Southern Barasano. For the latter, Jones & Jones (1991) presented a syntax monograph that has been reviewed by Dryer (1994). He criticises their type assignment[20] and concludes: "*It is possible that it is best treated as indeterminately SOV/OVS, a word order type that appears to be quite common in the Amazon basin.* (Dryer 1994: 63). Hixkaryána will be discussed together with the following set of languages.

In WALS[21] (Dryer & Haspelmath 2013), the following languages are listed as "OVS". Four of

[19] Dixon (1994:138) classifies Macushi as ergative. Dixon (2010, vol.1: 73) criticizes Ethnologue: "Macushi [...] is given as OVS, despite the excellent grammar of this language specifying that the 'basic orders' are OVA (although AOV also occurs frequently) and SV."

[20] "*A count of all examples in the grammar shows both SV and VS order common, with SV slightly more common, though numbers of examples cited in a grammar is a poor source of data. [...] If we interpret the notion of an OVS language as referring to clauses with a noun object and a noun subject (the standard usage in word order typology), it is not clear that Barasano qualifies*." (Dryer 1994: 63).

[21] In the introduction to chapter 82 of WALS, Dryer (2013) writes: "*There are also languages* […] *in which the*

them are plainly ergative, namely Kuikúro, Macushi, Päri, and Tuvaluan.[22] Four are caseless (i.e. 'neutral' alignment) but with ergative properties: Asurini, Selknam,[23] Tiriyo,[24] Ungarinjin.[25] According to Primus (1995:1089), "*The Tupi-Guarani languages Asurini and Oiampi have ergative marking in dependent clauses*."

Four potential candidates of OVS languages have to be discussed in detail further, namely Kxoe, Cubeo, Urarina,[26] and Hixkaryána. For Kxoe, Fehn's (2015: 214) grammar of Ts'ixa (Kalahari Kxoe) is very explicit: "*There are three patterns available for transitive clauses: AOV, AVO and OAV, with the latter occurring less frequently than the other two. Although the dominant word order of the Khoe languages is thought to be AOV (cf. Heine 1976, Güldemann 2014), AVO is just as frequent*." The type-assignment in WALS exclusively follows Köhler (1981).[27] In sum, Kxoe does not seem to qualify as a reliable testimony of OVS. So, we are left with Cubeo, Urarina, and Hixkaryána.

As for Cubeo, according to the typological platform SAILS [Muysken et al. (2016)], "*the patterns in the order of frequency in the data* [are]$_{HH}$*: OVS, SVO, VSO, SOV, and (least common) VOS* (M&M 1999:142)." WALS, which refers to the same source, namely Morse & Maxwell (1999), lists Cubeo as a nom-acc OVS language[28] that is generally head-final (postpositions, Gen-N, V-neg) and of the nominative-accusative type, with passive. This is crucial information for the subsequent analysis, as Hixkaryána and Urarina also exhibit these characteristics.

The essential issue to be settled for the three languages is this: Are these languages head-initial or head-final? If their VP is head-final, [OV] is likely to be a constituent. If they are head-initial, [VA] is a constituent preceded by O. The latter case would make them [O[VA]] languages, with "O" being the structurally highest argument in the clause. What are the relevant facts?

Cubeo, Urarina, Hixkaryna, are post-positional as well as Gen-N. According to Dryer (2007: 69) "*the fact that the characteristics in other languages pattern with the order of object and verb would lead us to expect both OVS and OSV languages to pattern with SOV languages. In so far as we have evidence, this prediction seems to be true. For example, Hixkaryana is*

order can be described as Absolutive-Verb-Ergative: these languages are shown as OVS on Map 81A and as SV on this map. In fact, three of the six OVS languages shown on Map 81A are of this type: Päri (Nilotic; Sudan; Andersen 1988), Mangarrayi (Mangarrayi; northern Australia; Merlan 1982) and Ungarinjin (Wororan; north-western Australia; Rumsey 1982)."

[22] Besnier 1986: 245: "*Despite the word-order freedom exhibited by Tuvalan, there is a basic order, and this order is verb initial.*" Besnier (2000: xxiv): *"Case marking follows an ergative-absolutive pattern".*

[23] "*Selk'nam seems to be an ergative language as to word order and verbal marking. Nevertheless, case marking is still an issue that remains to be debated, since the data now available is not sufficient to determine the typological nature of the language, which appears to have been an S marking/A-O unmarked language till the beginning of the twentieth century.*" Rojas-Berscia (2014: 23).

[24] Rill (2017: 430): "*In the end, Tiriyó verb agreement is best analyzed as ergative in alignment*."

[25] Rumsey (1982:145) summarizes the "ordering norms": S precedes V, O precedes V, while A follows. This is exactly the order one expects to find if a language is an SVO language with ergative alignment.

[26] A text count based on 445 main clauses sampled from seven texts produced the following frequencies: 3% OVA and 4% AOV orders (Olawsky 2006: 653; 2007: 45). 93% are clauses with null subjects and/or null objects. For dependent clauses, Olawsky (2006: 658) reports 0,3% VA and 0,8% AV orders.

[27] Köhler's reliability has been questioned: "*He himself reduced the richness of Khwe cultural and linguistic expressions in his documentation by increasingly limiting field methods*." (Boden 2018: 142).

[28] WALS follows the maxim of classifying a language according to the *more frequent* pattern. Ethnologue classifies this language as SOV. Chacon (2012: 190) notes, however "*The overall syntax of Kubeo allows a very flexible word order*." The reliable assignment to a word order type does not seem to be definitive.

postpositional and GN." The same is true for Urarina. In addition, as Kalin (2014: 1096) emphasizes, the adjective phrase is head-final, too. Olawsky (2006: 667-668) provides information on the V+Aux order of Urarina, an order that is completely absent in V-initial languages. Finally, Olawsky (2006: 662) notes that in negated sentences, AOV is an unmarked order, that is, A is not focussed. "*In a transitive clause, constituent order can be AOV as the result of negation*." Taken together, these grammatical features are good indicators for a head-final organization of the verb phrase in the three languages.

The cumulative evidence for a head-final VP has led Kalin (2014) to the conclusion, that Hixkaryana is an [[OV]S] language, with the VP[29] in a secondary, that is, fronted position. The same analysis, modulo head-initial V, is the standard structure of Malagasy, a head-initial language with VOS order. Its clause structure is taken to be [[V O] S], that is, with a clause-initial VP. (Paul 1999, Pearson 2001). Keenan & Manorohanta (2004: 178) formulate it as follows: S = [[Pred V°(Θ) + (X)] + DP].

This supports Derbyshire's (1981) original conjecture that the OVS serialisation pattern in Cariban languages is the echo of the change from ergative case marking to nom-acc. The structure [[OV]S ...] is the likely outcome when in an Abs-V-Erg system, case distinctions are reduced and the alignment mode is reinterpreted as nom-acc, while the word order patterns remain unchanged.[30] The price for maintaining the original order (13a) is greater structural complexity, namely a clause structure with a fronted VP (13a). In the other frequent order of Cariban languages, namely SOV (13b), the change has no effect on serialisation. This analysis covers the eight Cariban languages which Derbyshire & Pullum (1979) list as OVS languages.[31]

(13) a. $[P_{abs}\ [V°\ A_{erg}]] \Rightarrow\ [P_{acc}\ V°\ A_{nom}] = [[O_{acc}\ V°]_{VP}\ S_{nom}] = [[OV]S]$ (P = patient, A = agent)
b. $[P_{abs}\ [A_{erg}\ V°]] \Rightarrow\ [P_{acc}\ [A_{nom}\ [\text{--}\ V°]]] / [A_{nom}\ [P_{acc} V°]] =$ OSV / SOV

The resulting structure in (13a) corresponds to the analysis of Kalin (2014) for Hixkaryana. What we see in these Cariban languages is a rare constellation of clause structure, with a clause-initial head-final predicate phrase followed by the subject. The serialisation is O-V-S, but this is not an [O[VS]] clause structure and therefore not a structural counterpart to [S[VO]].

Strong support for an analysis with a clause-initial VP comes from Queixalós (2010: 241, 254) and a language from a different family. He argues that the clause structure of Katukína-Kanamarí is basically [[OV] S], and shows in detail that this structure even remains constant under *alternative alignments*. The frequent clause type is [[Erg V] Abs][32] and a less frequent one is [[Acc V] Nom]. Note that the arguments of the verb with dependent case change places in the two clause types, but the structure of the clauses remains constant in terms of the formal grammatical functions, namely [[OV]S]. In the traditional terminology, one of the two clause structures would wrongly be classified as ergative "SVO", and the other also wrongly as accusative

[29] "*Transitive clauses have a tightly bound OV verb phrase constituent that is usually followed by the subject NP. Des had actually said so in a dense 1961 paper I had not seen (IJAL 27, 125-142), packed with obscure formulae.*" (Geoffrey Pullum, Obituary: Desmond Derbyshire, *Linguist List* 19.1, Jan 03 2008).

[30] This is a well-known phenomenon in grammatical change. The old and the new grammar variant are weakly equivalent, with the *same* linearisation, but *differently* organised phrase structure.

[31] Apalaí, Arekuna-Taulipang, Hixkaryana, Macushi; Asuriní, Bacairí, Hianacoto-Umau, Panare. The first mentioned four languages still show morphological markers of ergativity.

[32] Note that this clause structure is a structure compatible with a clause-initial, preverbal position of an ergative. Here, the ergative is not in a preverbal subject position.

"OVS". The syntactically correct classification is [[$_{VP}$ XP V°] S]. All in all, there is no language known whose sentence structure would undoubtedly be an example of [O[VS]], which would be the structure of a genuine *structural* OVS language.

As for ergative languages, the traditional "OVS" classification means Absolutive-V-Ergative order, and this is *subject-verb-object* order under ergative alignment. It is a consequence of the above discussion that the *syntactical* identification of syntactical relations is an indispensable basis for a cross-linguistically adequate comparison of clause structures.

5. Closing remarks

The theory just presented is a grammar-centred theory of ergativity. The traditional theory, on the other hand, is communication-centred, which is characteristic of functionalist approaches. The basic difference becomes clear in the answer to the following question. Do communicative functions determine the range of forms[33] (= functionalism), or does the range of forms determine what is available for communicative functions (= structuralism)? Grammatical forms are not *caused* by communicative functions; rather, grammatical forms are *used* for communicative functions. And what is it that has 'coordinated' these two areas so far and continues to 'coordinate' them, viz. forms and functions? The cognitive evolution of grammars. The functional design of grammars is the result of hundreds of thousands of years of cognitive evolution of grammars; see Haider (2015b, 2021b, 2025) for details. Random variation and constant but 'blind' selection are two mechanisms of evolution which have led and still lead to a variety of *adaptive* designs in the absence of a designer. Grammars are not designed; they evolve.

What evolutionary quality of grammars lies behind ergativity? Why do we find more than one type of alignment across languages? The answer is once again the cognitive evolution of grammars. If the system potential leaves room for more than one outcome – see choice of the default in the alignment parameter – there will be more than one outcome in the long-run diachronic developments, cross-linguistically. Evolution is a dynamic process that opens up possibilities, not one that suppresses them. Which ones will eventually spread is an orthogonal question.

The step preceding the emergence of ergativity (and accusativity) is the emergence of order relations in terms of *dependent*-case, with structurally conditioned cases and a dependency relation between them. This is also the bifurcation point, namely the choice of the candidate for default case. It means that automatically, the other structurally case-marked argument receives the dependent case. What determines the choice of default and thereby the superordinate case?

It is a matter of chance and necessity. If the higher argument in the lexical argument ranking 'wins', an accusative system is born. If, however, the argument with higher informational salience[34] "wins" in the evolutionary 'competition' of grammar variants, we are at the bottom of the argument scale of the lexical argument structure, and the result is an abs-erg system. Both options are grammatically realistic, but the ergative option leads to a system with opposing hierarchies. The selection of the subject in an ergative system starts at the bottom (of the thematic hierarchy), but other syntactic relations typically work in a higher-governs-lower manner (cf. c-command relation). This inherent tension seems to be the main reason for the statistically

[33] „Range of forms" is the abbreviation for morphosyntactic inventory, syntactic structures, and relations.

[34] The default focus position of a linguistic message is typically the most deeply embedded core-argument, that is, at the end of a sentence. This is typically also the position of the undergoer argument of a transitive verb.

unequal distribution of ergative and accusative languages, and the reason for the so-called splits.

Evolution in general and the cognitive evolution of grammars in particular is not a strategically planning engineer but rather a tinkerer, as Jacob (1977) put it. This means that ergativity can also come into being by accident in a nom-acc language. A case in point is the acquired ergativity of a group of Indo-Aryan languages. They are nom-acc, except in the perfective tenses, where they are coding ergatively. How did this happen?

In most Indo-European languages, the tense systems have gradually changed to a so-called analytical system, which typically consists of an uninflected verb form plus an inflected auxiliary verb. In Romance and Germanic languages, for instance, the perfect is coded with a participle plus an auxiliary. For all verbs (except unaccusative ones in languages with auxiliary selection) the auxiliary with which they are combined is typically either a former transitive verb of possession (*habere, avere, avoir*, ...) or a get-into-possession verb as e.g. the Germanic *have*, a cognate of Latin *capere* (grasp). If, on the other hand, the participle is combined with the available *copula*, the result is an unavoidable passive effect since the past participle form generally blocks the default subject argument. The '*have*'-type auxiliaries deblock; see Haider (1984), Haider & Rindler-Schjerve (1987). The copula-type auxiliaries, unlike auxiliaries of the '*have*'-kind, do not deblock a blocked subject argument.[35] And that is the point at which the Indo-Aryan languages branched off. In the ergative group, no deblocking auxiliary has been recruited. So, they had to make do with the copula.

As a consequence, the perfect tense worked like a passivising construction, with the default subject of the base verb in an oblique case, although, as a tense construction, it was part of otherwise *active* constructions. Within a few centuries, this got streamlined into an ergatively-coding tense paradigm, with the direct object of the non-perfective tensed variants as grammatical subject in the perfective tenses, viz. absolutive.

Please note once again that there can be no doubt that in Indo-Aryan ergative constructions, the absolutive is the case of the grammatical subject. This is a consequence of its origin in an originally passivising construction. The Indo-Aryan ergative case derives from an oblique case that is typical for Indo-European passive constructions, i.e. from instrumental (or from ablative). The subsequent integration into the grammatical system follows familiar patterns, as the ergative paradigm is limited to perfect tenses, while finite verbs in the other tenses are always used in the nom-acc mode in Indo-Aryan. The so-called tense split is an accident in the course of grammatical changes in the tense-coding system in these languages.

Summary

The grammatical elegance of an ergative language becomes evident when it is not read against the grain. Once one has understood why the absolutive is the case of the *grammatical* subject, facts fall in place and and the previous apparent disorder gives way to a well-organized

[35] Here is an example set from German; see Haider & Schjerve (1987):

i. Der Junge *hat* ihm den Brief *zugestellt* – der *zugestellte* Brief – *der *zugestellte* Junge – der *zustellende* Junge
the boy has delivered him the letter – the delivered letter – *the delivered boy – the delivering boy

ii. Der Brief *wurde* ihm (von einem Jungen) *zugestellt* (passive) – Der Brief *ist zugestellt* (copula + adject. part.)
the letter was him (by a boy) delivered – the letter is delivered

grammatical architecture, whereby the grammatical parallels between the two types of alignment become plainly visible.

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